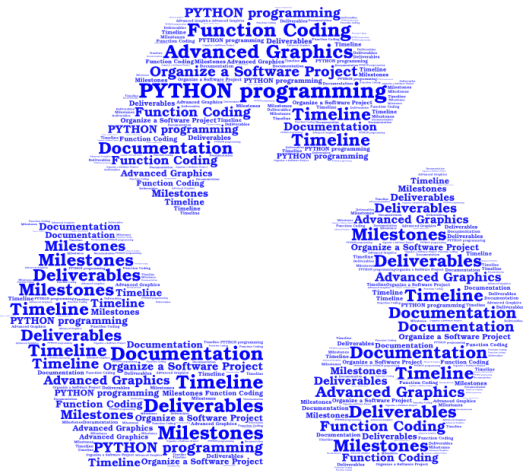




Python Programming: First Steps

Lecture: Modelling an Simulation

This is a compulsory elective module
of
**Technology of Circular
Economy**
(Master of Engineering)

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What is programming?

Telling a computer what to do. Example: „Alexa, play some music!“

Actually, that is a command.

A program would be a sequence of commands, like:

„Alexa, play a song from Linkin Park, Adele and Otto Walkes“

Lets look at this line:

command = „play“

data = „ Linkin Park, Adele and Otto Walkes“ (ordered list of strings)

What is programming?

Translated program:

1. play song from „Linkin Park“
2. play song from „Adele“
3. play song from „Otto Walkes“

<https://www.youtube.com/watch?v=GgcJitRTn38>

Programming and data types which you already know from using MS EXCEL

EXCEL

All data is arranged in „cells“ = containers
„Cells“ can also contain equations (small programs)
How does EXCEL know what is what? (cell format)

working with EXCEL

All data is visible during the whole time of working.

EXCEL is great, but there are also a number of limits coming with this approach.

E.g. maximum number of characters in cell = 255

Think about large data structures, complicated equations, ...



Why PYTHON?

Python was first released in 1991. The name of the Python programming language comes from an old BBC television comedy sketch series called *Monty Python's Flying Circus*.

-> Python is 'funny'.

... and you can make jokes about it, like:

Don't let Python bite you!



Why PYTHON?

Python was first released in 1991. The name of the Python programming language comes from an old BBC television comedy sketch series called *Monty Python's Flying Circus*.

-> Python is 'funny'.

Why using Python?

- It is easy to learn.
- It is easy to use.
- It is easy to obtain, install and deploy – Python is free, open and multiplatform.

Python is:

- very flexible
- an 'intelligent' language – many things happen in the background

discuss: interpreter vs compiler language



Free resources for learning Python:

<https://www.python.org/> (Tutorial and documentation)

<https://www.w3schools.com/python/default.asp> (Tutorial and documentation)

<https://www.learnpython.org/> (Tutorial)

<https://www.geeksforgeeks.org/python/python-programming-language-tutorial/> (Tutorial)

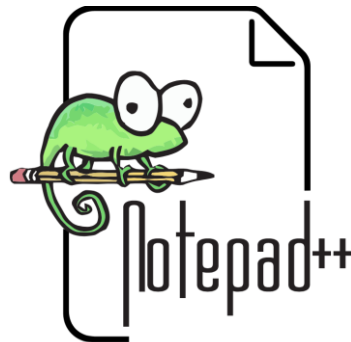
... and many other resources.



We will be doing it the minimalistic way. No special computers required, even 10 year old computers will work.

What do you need?

- Python installed on your computer. (Works also online, but with low performance.)
- A simple text editor with syntax highlighting. (We use Notepad++ but many others will work too.)
- A console is optional, but you can run programs also by mouse click.



```
PS C:\Users\uw5002> ls

Verzeichnis: C:\Users\uw5002

Mode                LastWriteTime         Length Name
----                -
d-----         25.10.2024    10:16         .android
d-----        16.01.2025     09:57       .matplotlib
d-----         05.08.2024     09:13       .ms-ad
d-r-----         05.08.2024     08:33       3D Objects
d-r-----         05.08.2024     08:33       Contacts
d-r-----        27.08.2025     09:33       Desktop
d-r-----        17.09.2025     08:47       Documents
d-r-----        25.09.2025     12:46       Downloads
d-r-----         05.08.2024     08:33       Favorites
d-r-----         05.08.2024     08:33       Links
d-r-----         05.08.2024     08:33       Music
d-r-----        29.04.2025     10:17       Pictures
d-r-----         05.08.2024     08:33       Saved Games
d-r-----         05.08.2024     08:34       Searches
d-r-----        10.01.2025     13:01       Videos

PS C:\Users\uw5002>
```

There is nothing to pay extra for. Everything is already on your computer or it is free to download.

Create folder where you want to store your Python files on the local computer.

Open Notepad++

Create a new file, save as "test-01.py".

Type in the short program.

Now, you have 3 options.



```
1 a = 'Hello World!'
2 b = 1 + 4
3 print(a)
4 print(b)
```

1

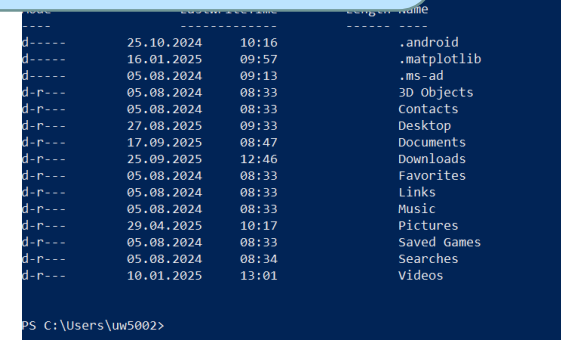
Run Python program in a web-interface
(cloud service provided by the Academic Cloud)
Jupyter

2

Run Python program from a file by double-clicking
the icon with the mouse. (batch file)

3

Run Python program from the command
line (in Windows = PowerShell).



1

Go to the Academic Cloud, Login, start Jupyter (with standard notebook).

Type in your example program.

Run the program (either by the menu 'Run' or by hitting Shift + Enter).

```
[2]: a = 'Hello World!'  
     b = 1 + 4  
     print(a)  
     print(b)
```

```
Hello World!  
5
```

Windows recognizes files with the extension *.bat as being executable (batch files).

Usually, they contain a command that can be recognized by the operating system of the computer.

It can be a single command or a list of commands that are executed one after each other as in a batch (=> batch file).

Now:

- 1) Create a new text file in Notepad++ and save as 'myrun.bat' in the same directory as your python file.
- 2) Type the following command: `python test-01.py > out.txt`
- 3) Save the file.
- 4) Double click with the mouse on the file 'myrun.bat'.
- 5) The file 'out.txt' is created -> open it in the text editor.

test-01.py

```
1 a = 'Hello World! '  
2 b = 1 + 4  
3 print(a)  
4 print(b)
```

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Now:

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- 3) Save the file.
- 4) Double click with the mouse on the file 'myrun.bat'.
- 5) The file 'out.txt' is created -> open it in the text editor.

This is very easy to use.

Disadvantage: if something goes wrong, you do not get any information.

Lets modify our program to do something wrong ->

test-01.py

```
1 a = 'Hello World!'  
2 b = 1 + 4  
3 print(a)  
4 print(b)  
5  
6 c = a + b  
7 print(c)
```

Running a program – PowerShell

Run the PowerShell from the Windows Start menu.

Navigate with text command 'cd' to your directory containing the Python file.

useful commands

dir	= print current directory
cd	= change directory
ls	= list files and directories
clear	= clear the screen

useful keys



auto complete (if you type half the name ...)



history (show last command, you can do it several times)

Now, type 'python test-01.py' and hit enter key. There should be an error message.

Python is an object oriented language.

object, e.g. a number „1“

variables can be understood as references to the object

`x = 1`

Data types:

Data types in Python are a way to classify data items.

They represent the kind of value that you can store, which in turn determines what operations can be performed on that data.

PYTHON – basic data types

Numbers:

int

```
x = 42
```

float

```
x = 3.1427
```

complex

```
x = 1j
```

Examples:

https://www.w3schools.com/python/python_datatypes.asp

Sequences and collections:

String

A string is a sequence of characters.

```
x = 'Circular Economy'
```

List

A list is a sequence of things.

```
x = [ 1, 42, 6, 9 ]  
y = [ 1, 42, 'Circular Economy' ]
```

Dictionary

A dictionary is a collection of things.

```
x = [ 'me': 2, 'you': 11 ]
```

Tuple

A tuple is a list that can not be changed.

```
x = ( 1, 42, 6, 9 )  
y = ( 1, 42, 'Circular Economy' )
```

Fundamental:

None

```
x = None
```

Boolean

```
x = True  
y = False
```

Data types are important. Python can do some 'magic' in the background.

Example: what is $1 + 1 = ?$

Data types are important. Python can do some 'magic' in the background.

Example: what is $1 + 1 = ?$

2 or 11

Example: what is $3 * 1 = ?$

Data types are important. Python can do some 'magic' in the background.

Example: what is $1+1=$?

2 or 11

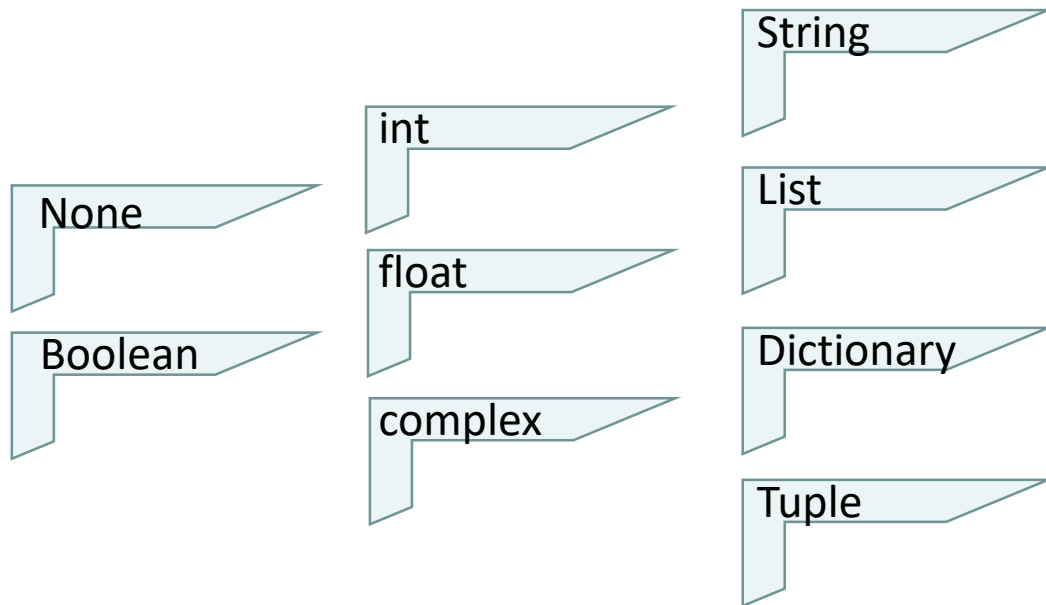
Example: what is $3 * 1 =$?

3 or 111

If you have different data types, than $3 * 1$ is NOT equal to $1 * 3$!

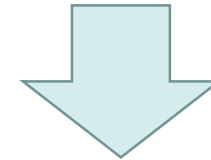
Short summary - data types

Remember: PYTHON – basic data types



Data types:

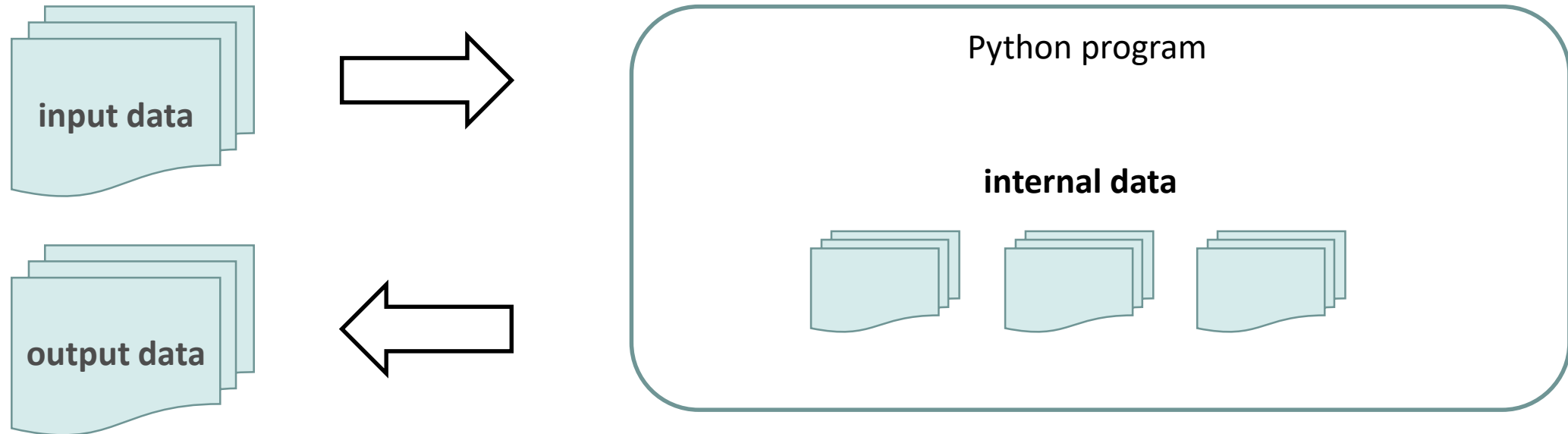
Data types in Python are a way to classify data items. They represent the kind of value, which determines what operations can be performed on that data.



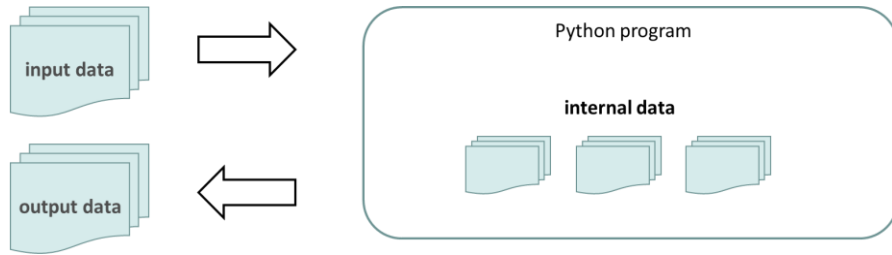
Why is this important?

Because you need to choose the right data type for your task.

Choosing the right type means to create a data model.



The data model



You need to work with the input data.
You need to create the output data.

The internal data structure must be suitable to do so.

You can choose from the existing data types (Python built in) or create new ones by combining the basic types.

For small programs that is a no brainer.

For large software projects this is an essential part, which has major impact on the flexibility and performance of the final software.

At the end it all boils down to: THINK - before you start to program larger things!

What you should have learned today.

- Where to find information about Python in the internet.
- How to run simple Python programs, using
 - a text editor
 - a command shell (command line)
 - a batch file
- What are the main data structures in Python and why you should care.